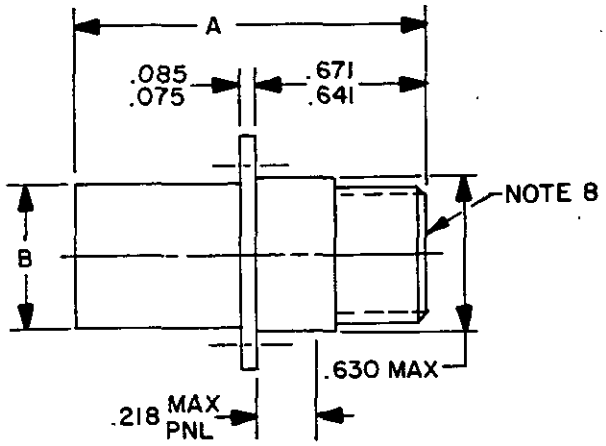
MILITARY SPECIFICATION SHEET

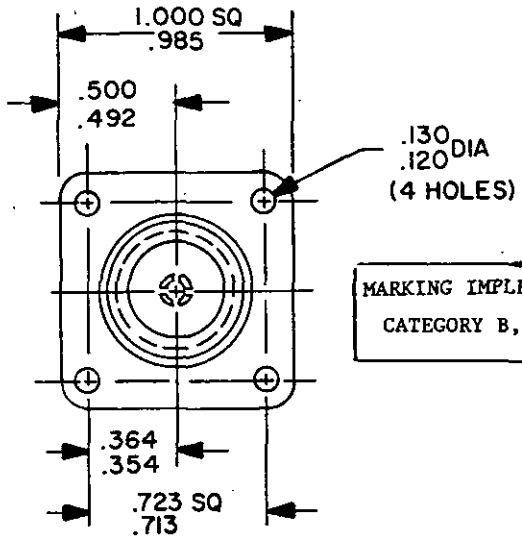
CONNECTORS, PLUGS AND RECEPTACLES, ELECTRICAL, COAXIAL, RADIO FREQUENCY, (SERIES N (CABLED), FLANGE MOUNTED, SOCKET CONTACT, CLASS 2)

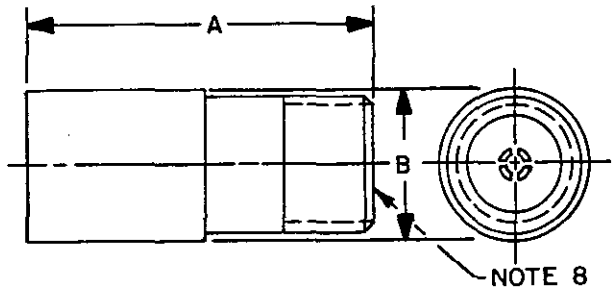
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.



RECEPTACLE





PLUG

Inches	mm	Inches	mm
.075	1.91	.500	12.70
.085	2.16	.630	16.00
.120	3.05	.641	16.28
.130	3.30	.671	17.04
.218	5.54	.713	18.11
.354	8.99	.723	18.36
.364	9.25	.985	25.02
.492	12.50	1.005	25.53

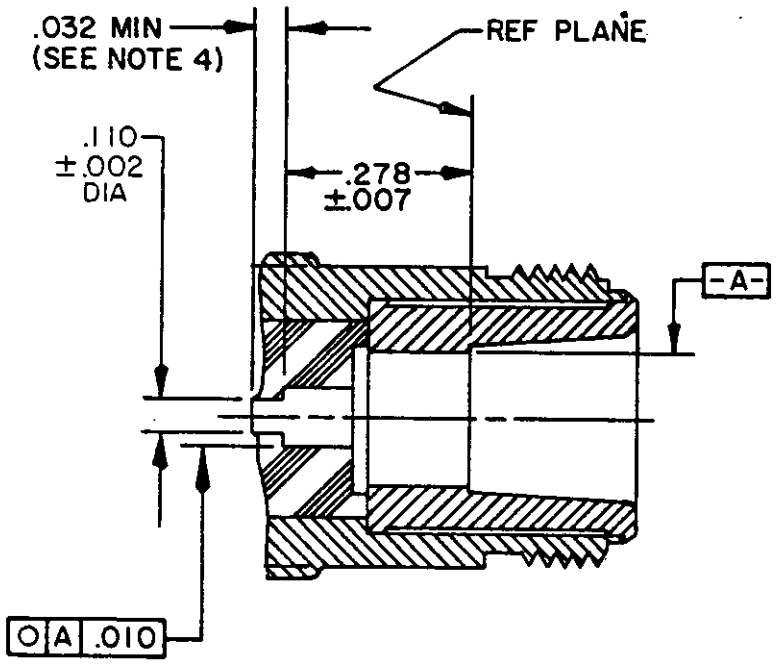
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. See table I and III for dimensions A and B.
4. Dimensions A and B are the largest overall dimensions of the connector less the flange.
5. Wrench flats are to accommodate standard wrench opening in accordance with FED-STD-H28, appendix 10.
6. All undimensioned pictorial representations are for reference purposes only.
7. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
8. Series N, socket contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.

AMSC N/A
FSC 5935

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CATEGORY D

Inches	mm		Inches	mm
.002	1.91		.032	0.81
.007	2.16		.110	2.79
.010	3.05			

NOTES:
1. Dimensions are in inches.
2. Slot design optional.
3. Metric equivalents are given for general information only.
4. Chamfer is optional, if chamfer is used put chamfer on a 30° maximum.

FIGURE 2. Category D captivation detail.

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TABLE I. Dash numbers, cross reference, and dimensions.

Dash number 1/	Type	Applicable cable M17/ #	Dimensions	Inches (millimeters) maximum
Category A - Field serviceable (no special tools required)				
0101 4/	Plug	2-RG6Δ 073-RG212* RG-222/U		
0003 4/	Plug	6-RG11Δ 62-RG144Δ 065-RG165 074-RG213 75-RG214* 77-RG216Δ RG-225/U@	A B	1.672(42.47) .781(19.84)
0131 4/	Plug	074-RG215*†	A B	2.350(59.69) .781(19.84)
0104 4/	Receptacle	2-RG6Δ 073-RG212* RG-222/U		
0006 4/	Receptacle	6-RG11Δ 62-RG144Δ 065-RG165 074-RG213 75-RG214* 77-RG216Δ RG-225/U@	A B	1.672(42.47) .750(19.05)
0132 4/	Receptacle	074-RG215†	A B	2.350(59.69) .781(19.84)
0133 4/		078-RG217	A B	1.781(44.84) .906(23.01)
Category C - Field replaceable (MIL-C-22520 crimp tool) 3/ See footnote symbol next to applicable cable for crimp die				
0019 4/	Plug	073-RG212*≠ RG-222/U≠		
0020 4/		074-RG213*≮	A B	1.938(49.23) .781(19.84)
0021 4/		75-RG214*≮		
0024 4/		065-RG165*@≮		
0025 4/		RG-225/U@≮	A B	1.938(49.23) .781(19.84)
0026 4/		6-RG11Δ≮ 62-RG144Δ@*≮		
0039 4/		77-RG216*Δ≮		
0040 4/		2-RG6*Δ≠		

See footnotes at end of table.

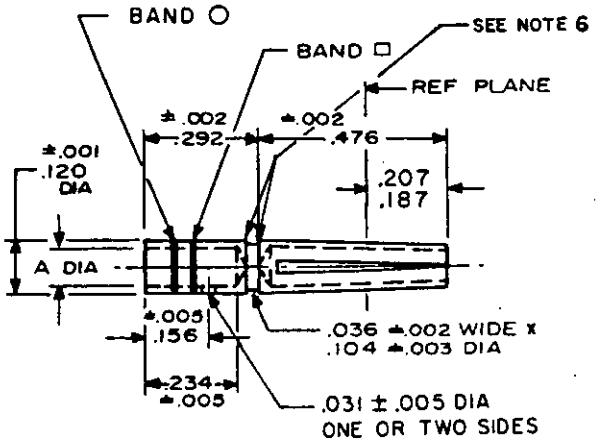
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:38Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers, cross reference, and dimensions - Continued.

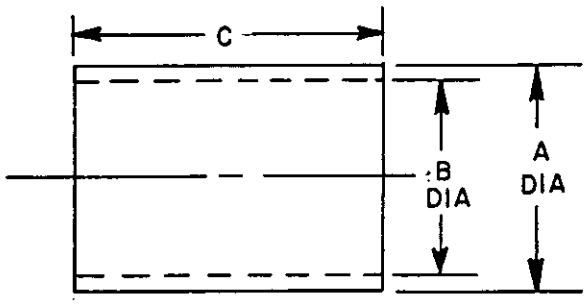
Dash number 1/	Type	Applicable cable M17/ #	Dimensions	Inches (millimeters) maximum
Category C - Field replaceable (MIL-C-22520 crimp tool) 3/ See footnote symbol next to applicable cable for crimp die				
0027 4/	Receptacle	073-RG212*≠ RG-222/U≠		
0028 4/	Receptacle	074-RG213*≺		
0029 4/	Receptacle	75-RG214*≺		
0030 4/	Receptacle	77-RG216*Δ≺	A B	1.938(47.23) .750(19.05)
0034 4/	Receptacle	RG-225/U@≺		
0041 4/		065-RG165*@≺		
0042 4/		6-RG11Δ≺ 62-RG144Δ@*≺		
0043 4/		2-RG6*Δ≠		
Category D - Field replaceable - defined piece parts 3/5/2/				
0501 4/	Plug	75-RG214* RG-225/U 127-RG393	A B	1.656(42.07) .750(19.05)
0502 4/		074-RG213		
0503 4/		060-RG142@* 128-RG400	A B	1.593(40.48) .750(19.05)
0504 4/		028-RG058		
0511 4/	Receptacle	75-RG214* RG-225/U@ 127-RG393	A B	1.656(42.07) .750(19.05)
0512 4/	Receptacle	074-RG213*		
0513 4/	Receptacle	060-RG142*@ 128-RG400	A B	1.593(40.48) .750(19.05)
0514 4/		028-RG058		

1/ For cross reference of dash number to superseded part number or type designation, see table IV.
2/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.
3/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.
4/ These connectors have captivated center contacts.
5/ Not to be used in Army equipment.
The latest version of each cable shall be applicable.
* Cable to be used when performing tests requiring cable except as in notes @ and Δ.
@ Cable to be used for the +200°C temperature cycling tests.
Δ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.
+ Armored cable.
≠ M22520/5-35 Closure A or M22520/5-55 Closure A.
≺ M22520/5-61

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CENTER CONTACT



CRIMP FERRULE

Dash No.	Contact No. 2/	A	Basic crimp tool 1/ M22520/1	Crimp die or positioner M22520/1	Crimp tensile min	Color band □	Color band ○
0501 0502 0511 0512	2-10	.098 ±.002	-01	-13	60 lbs	Red	Brown
0504 0503 0513 0514	2-11	.043 +.001 -.002	-01	-13	20 lbs	Blue	Brown

Dash No.	Ferrule No. 2/	A	B	C ±.015	Basic crimp tool 1/ M22520/5	Crimp die or positioner M22520/5
0501 0511	2-50	.492 +.003 -.000	.438 +.003 -.000	.600	-01	-25 Closure A or -61
0502 0512	2-51		.418 +.003 -.000			
0503 0513	2-52	.250 ±.003	.220 ±.003	.500		-5, -11 or 57 Closure A or -19 Closure B
0504 0514	2-53	.245 ±.001	.206 ±.001	.500		

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

FIGURE 3. Contact and ferrule dimensions for category D only.

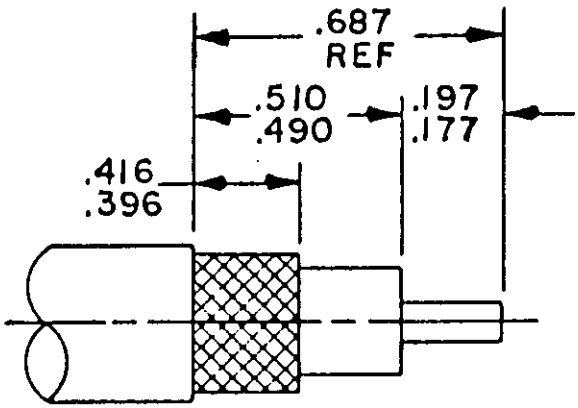
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Inches	mm	Inches	mm
.001	0.03	.187	4.75
.002	0.05	.207	5.26
.003	0.08	.220	5.59
.005	0.13	.234	5.94
.015	0.38	.250	6.35
.031	0.79	.292	7.42
.036	0.91	.418	10.62
.043	1.09	.438	11.13
.098	2.49	.476	12.09
.104	2.64	.492	12.50
.120	3.05	.500	12.70
.156	3.96	.600	15.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Center contact material shall be phosphor bronze or copper beryllium.
4. Crimp tensile test shall be in accordance with MIL-C-39029.
5. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
6. .003 (0.08 mm) maximum break.

FIGURE 3. Contact and ferrule dimensions for category D only - Continued.

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Inches	mm
.177	4.50
.197	5.00
.396	10.06
.416	10.57
.490	12.45
.510	12.95
.687	17.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Recommended cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:
1,000 volts rms, maximum working voltage at sea level.

250 volts rms, maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 6 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque, 6 to 10 inch-pounds.

Mating characteristics:

Reference MIL-STD-348 and figure 2 for dimensions.

Center contact (socket):

Oversize test pin: 0.074 diameter, minimum (non-closed entry contacts only).

Insertion depth: .125, minimum.

Number of insertions: One.

Insertion force test: Steel test pin diameter .066, minimum.

Test pin finish: 16 microinches.

Insertion force: 2 pounds, maximum.

Withdrawal force test: Steel test pin diameter .063, maximum.

Withdrawal force: 2 ounces, minimum.

Test pin finish: 16 microinches.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B, 5,000 megohms, minimum.

Center contact retention: (Applicable to captivated-center-contact connectors only). Fifteen pounds, minimum axial force for all cable except RG400 and RG142; 6 lbs, minimum for RG400 and RG142.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30, maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:38Z
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Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.008+.002 F (F in GHz).

Item 16: VSWR shall be less than 1.008+.002 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.012+.004 F (F in GHz).

Group B inspection: VSWR shall be less than 1.024+.007 F (F in GHz).

Qualification and group C inspection: VSWR shall not exceed 1.08.

Connector durability: 500 cycles, minimum at 12 cycles per minute, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum:

	Initial	After environment
Center contact	1.0	1.5
Outer contact	.2	Not applicable
Braid to body	.05	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202, 2,500 volts rms, minimum at sea level for connectors using other than RG400 and RG142; 1,500 volts rms, minimum for connectors using RG400 and RG142.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see tables I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage: 500 volts, minimum.

Altitude: 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency: 1,500 volts rms tested at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:38Z
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Cable retention force:

Noncrimp assemblies: 75 pounds, minimum.

Crimp assemblies:

50 pounds, minimum for cables .155 - .189 OD.

60 pounds, minimum for cables .190 - .229 OD.

75 pounds, minimum for cables .230 - .249 OD.

90 pounds, minimum for cables .250 OD and larger.

Coupling mechanism retention force: Not applicable.

RF leakage: -90 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss: .15 dB maximum, tested at 10 GHz.

Part number: M39012/02 (dash number from table I or "B" number from table III).

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TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/	M39012/
I	02-0101 02-0003 02-0104 03-0101	02-0101 02-0003 02-0104 03-0101
II	02-0003 02-0006 03-0012	02-0003 02-0006 03-0012
III	02B0007 02B0008 02B0009 02B0012 02B0013 02B0015 02B0016 02B0017 03B0003 03B0004 03B0005 03B0008 03B0009	02B0007 02B0008 02B0009 02B0012 02B0013 02B0014 02B0015 02B0016 02B0017 02B0018 03B0003 03B0004 03B0005 03B0008 03B0009 03B0010 03B0011
IV	02-0019 02-0020 02-0021 02-0024 02-0025 02-0027 02-0028 02-0029 02-0034 03-0013 03-0014 03-0015 03-0016 03-0018 03-0019	02B0007 02B0008 02B0009 02B0012 02B0013 02B0014 02B0015 02B0016 02B0017 02B0018 02-0019 02-0020 02-0021 02-0024 02-0025 02-0026 02-0027 02-0028 02-0029 02-0030 02-0034 03B0003 03B0004 03B0005 03B0008 03B0009 03B0010 03B0011 03-0013 03-0014 03-0015 03-0016

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TABLE II. Group qualification - Continued.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/	M39012/
IV		03-0018 03-0019 03-0020 03-0021
	M39012/	M39012/
V	02B0014 02B0018 03B0010 03B0011	02B0014 02B0018 03B0010 03B0011
VI	02-0026 02-0030 02-0039 02-0040 02-0042 02-0043 03-0020 03-0021	02B0014 02B0018 02-0026 02-0030 02-0039 02-0040 02-0042 02-0043 03B0010 03B0011 03-0020 03-0021
VII	02-0131 02-0132 02-0133	02-0131 02-0132 02-0133
VIII	02-0034 02B0036 02B0037 02B0038 02B0044	02-0034 02B0036 02B0037 02B0038 02B0044
IX	02B0035	02B0035
X	02-0501 02-0502 02-0511 02-0512	02-0501 02-0502 02-0503 02-0511 02-0512 02-0513
XI	02-0503 02-0513	02-0503 02-0513
XII	02-0504 02-0514	02-0504 02-0514

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE III. Category B - Nonfield replaceable (special tools may be required).

Not for Air Force or Navy use. For OEM use only.

Part number 1/ 3/	Type	Applicable cable M17/ #	Dimensions	Inches (millimeters) maximum
M39012/02B0007 4/	Plug	073-RG212* RG-222/U		
M39012/02B0008 4/		074-RG213*	A	1.938(49.23)
M39012/02B0009 4/		75-RG214*	B	.781(19.84)
M39012/02B0012 4/		65-RG165*@		
M39012/02B0013 4/		RG-225/U*@		
M39012/02B0014 4/		62-RG144*Δ@ 6-RG11Δ	A	1.938(49.23)
M39012/02B0018 4/		77-RG216*Δ	B	.781(19.84)
M39012/02B0044 2/		2-RG6*Δ		
M39012/02B0015 4/	Receptacle	073-RG212* RG-222/U		
M39012/02B0016 4/	Receptacle	074-RG213*		
M39012/02B0017 4/		75-RG214*	A	1.938(49.23)
M39012/02B0035 2/		065-RG165*@	B	.750(19.05)
M39012/02B0036 2/		6-RG11Δ 62-RG144Δ*@		
M39012/02B0037 2/		77-RG216*Δ	A	1.938(49.23)
M39012/02B0038 2/		2-RG6*Δ	B	.750(19.05)

1/ For cross reference of dash number to superseded part number or type designation, see table IV.
2/ These connectors have captivated center contacts.
3/ For maintenance replacements for category B, see table V.
4/ Inactive for new design.
The latest version of each cable shall be applicable.
* Cable to be used when performing tests requiring cable except as in notes @ and Δ.
@ Cable to be used for the +200°C temperature cycling tests.
Δ These are not 50 ohm cables; therefore, when attached to the specified connectors, VSWR, RF leakage, and insertion loss are not applicable.

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TABLE IV. Cross reference of part numbers.

Preferred part number M39012/02	Substitute for part number or type designation 1/2/
-0101	MS91233-20D, M39012/02-0001
-0003	MS91237-23F
-0003	MS91235-1186A, M39012/02-0002
-0104	MS91233-19E, M39012/02-0004
-0006	MS91237-22F
-0006	MS90536-1187A, M39012/02-0005
B0007 3/	M23329/2-11, UG-1687/U, M39012/02-0011, UG-1691/U, M39012/02-0007
B0008 3/	M23329/2-12, UG-1688/U, M39012/02-0008
B0009 3/	M23329/2-13, UG-1689/U, M39012/02-0009
B0012 3/	M23329/2-17, UG-1692/U, M39012/02-0012
B0013 3/	M23329/2-18, UG-1693/U, M39012/02-0013
B0014 3/	M23329/2-19, UG-1694/U, M39012/02-0014
B0015 3/	UG-1696/U, M39012/02-0015
B0016 3/	UG-1697/U, M39012/02-0016
B0017 3/	UG-1698/U, M39012/02-0017
B0018 3/	UG-1818/U, M39012/02-0018
-0019	M39012/02-0023
-0020	
-0021	
-0024	
-0025	
-0026	
-0027	
-0028	
-0029	
-0030	
-0131	UG-940C/U, M39012/02-0031
-0132	UG-935D/U, M39012/02-0032
-0133	
-0034	
B0035 3/	M39012/02-0035
B0036 3/	M39012/02-0036
B0037 3/	M39012/02-0037
B0038 3/	M39012/02-0038
-0039	
-0040	
-0041	
-0042	
-0043	
B0044	M39012/02-0044
-0501	
-0502	
-0503	
-0504	
-0511	M39012/02-0033
-0512	
-0513	
-0514	UG1095/U

1/ The superseded part number or the type designation is for cross reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/02-XXXX shall be used in all cases for marking and identifying the connector.

2/ The basic type designation includes all letter versions of the specified number, e.g. UG-18/U includes UG-18 A/U, UG-18B/U, etc.

3/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

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TABLE V. Maintenance replacements for category B.

Category B number*	Category C dash number	Category A dash number	Category D dash number
Plugs:			
B0007	0019	0101	---
B0008	0020	0003	0502
B0009	0021	0003	0501
B0012	0024	0003	---
B0013	0025	0003	0501
B0014	0026	0003	---
B0044	0040	0101	---
Receptacles:			
B0015	0027	0104	---
B0016	0028	0006	0512
B0017	0029	0006	0511
B0033	0034	0006	0511
B0035	0041	0006	---
B0036	0042	0006	---
B0037	0030	0006	---
B0038	0043	0104	---

*Category B connectors are for original installation only. They will not be stocked or acquired by the Government.

Revison letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - CR, EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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